

The spatial spillover effect and mechanism of carbon trading policy on
pollution reduction and carbon reduction: Evidence from the Pearl

River-Xijiang River Economic Belt in China

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The spatial spillover effect and mechanism of carbon trading policy on pollution reduction and carbon reduction: Evidence from the Pearl River-Xijiang River Economic Belt in China

Abstract

The research explores the effects of Carbon Trading Policy (CTP) on pollution and carbon reduction in the Pearl River-Xijiang Economic Belt (2008-2021) using SCM and S-DID models. The results show a 34.1% overall reduction in pollution and carbon levels, with a 29.9% decrease via spatial spillovers. CTP is most effective in reducing CO₂ and SO₂, while less effective on PM_{2.5}. Spillover effects on technological innovation surpass direct impacts, but the policy has a negative effect on industrial structure. We recommend improving coordinated governance and tailoring pollution and carbon reduction strategies to local conditions.

Keywords: carbon trading policy pollution reduction and carbon reduction
synthesis control spatial spillover effect mechanism of action

1.Introduction

Climate warming, environmental pollution, and energy crises are challenges faced globally. CTP, as a market-oriented means of environmental regulation, have become one of the key strategies for governments to deal with global warming and achieve carbon emission reduction targets[1]. Since the end of the 20th century, CTP have been established around the world. Through the introduction of the trading mechanism of emission rights, enterprises are encouraged to seek low-carbon development paths, thus promoting the sustainable development of the whole society[1]. China, as one of the world's largest carbon-emitting countries, has made remarkable progress in the construction of the carbon trading market in recent years. As an important domestic economic region, the implementation effect and mechanism of carbon trading policies in the Zhuhai-Xijiang Economic Belt are of great significance for understanding the national and even global carbon emission reduction process. CTP is a market-based environmental policy tool designed to encourage companies to reduce their greenhouse gas emissions by establishing a market for carbon emission allowances[1]. It is a market-based approach to pollution control that incentivizes companies to reduce emissions by setting a cap on emissions for a certain period of time and allowing the buying and selling of emission rights within that cap[2]. Participants in the carbon trading market may include governments, businesses

and non-governmental organizations, which can participate in the market by buying or selling emission rights[3]. With the rapid progress of industrialization and urbanization in China, the energy demand continues to grow, and the task of reducing pollution is arduous. Given the "common source of carbon and pollution" and the close association between the "Beautiful China" initiative and the "dual carbon" goals, it is urgent to advance pollution and carbon reduction in a coordinated manner. China's 14th Five-Year Plan and the "Implementation Plan for Synergistic Efficiency in Pollution and Carbon Reduction" both emphasize the synergistic improvement of pollution and carbon reduction. However, between 2015 and 2019, only 31% of 335 prefecture-level and above cities in China achieved coordinated pollution and carbon reduction, prompting deep reflection on synergistic emission reductions **Error! Reference source not found..** In terms of regional coordination, taking carbon reduction actions while improving environmental quality and strengthening regional cooperation can enhance overall emission reduction effects. In terms of goal coordination, it is projected that by 2060, the average concentration of PM_{2.5} in China will reach 10 µg/m³, with 80% of this effort driven by "dual carbon" policies[4]. The "Opinions on Deepening the Battle Against Pollution" also indicates that advancing pollution and carbon reduction in a coordinated manner requires full use of market-based mechanisms. Among these, CTP which can control the total carbon emissions and is incentivizing, is an important means to achieve the "dual carbon" goals and is more suitable for China's long-term development[6].

CTP is based on the theories of externality, Coase theorem, and Dales' emission trading theory, internalizing external costs and directly or indirectly achieving carbon reduction. It has garnered significant attention due to its flexibility, low post-implementation reduction costs, and overall effectiveness in emission reduction **Error! Reference source not found..** As the policy's implementation progresses, research on the CTP has evolved from qualitative theoretical analyses, such as mechanism design[7]. and policy comparison, to quantitative studies, including policy evaluation and mechanism analysis[10]. Early research focused on the singular impact of CTP on CO₂ or atmospheric pollutant reduction, reaching some consensus: CTP can significantly reduce carbon emissions[11] and carbon intensity[12] in pilot areas. However, due to differences in economic development levels, industrial structure, and policy execution efficiency, there is heterogeneity in the carbon reduction effects of pilot programs [13] attention, offering a new perspective for the formulation and evaluation of climate change and environmental

governance policies. The concept of synergistic effects was proposed by Ayres et al. and later introduced into the IPCC assessment reports, indicating that emission reduction measures often generate widespread positive impacts. Existing studies explore the synergistic emission reduction effects of CTP on pollution and carbon reduction using methods like panel regression and difference-in-differences (DID), mainly reflected in the synergy between SO₂ and CO₂, though there is regional heterogeneity[14]. It also involves the synergistic reduction of PM_{2.5}[15], smoke and dust[16], and solid waste [17], but consistent conclusions have yet to be reached. As research deepens, scholars have gradually begun to focus on the mechanisms of pollution and carbon reduction in carbon trading policies[18], including energy structure, industrial structure, and technological innovation.

As a key instrument in achieving the global "dual carbon" goals, CPT has gradually demonstrated its effectiveness in reducing pollution and carbon emissions in China. In economically developed regions such as the Yangtze River Delta[19], Beijing-Tianjin-Hebei [20], and the Pearl River Delta, the carbon trading system is relatively mature, and its policy is analyzed. In contrast, the Pearl River-West River Economic Belt exhibits significant differences in policy frameworks, industrial structure, and resource endowment, with economically developed cities like Guangzhou and resource-scarce areas such as the Guangxi West River basin[21]. These complex regional characteristics present challenges to the development and stability of clean energy supply, and synergies are significant. These complex regional characteristics present challenges to the development and stability of clean energy supply, while also exacerbating disparities in carbon and pollutant emissions, increasing the difficulty of coordinating regional emission reductions. The differences in economic development levels, energy structures, and policy responses among cities mean that CPT in the Pearl River-West River Economic Belt requires an integrated approach to coordinate policy support, technological innovation, and regional synergies. First, this region is at a critical stage of industrial transformation and economic coordination, urgently needing carbon trading policies to incentivize multi-level market mechanisms and policy levers to address the disconnect between clean energy supply and pollution and carbon reduction efforts. **Error! Reference source not found..** Second, the Pearl River-West River Economic Belt is not only a key region for achieving national "dual carbon" goals but also an ideal testing ground for studying the implementation effects of multi-level carbon trading policies, due to its unique regional characteristics. Unlike market-driven regions such as the Yangtze

River Delta and Beijing-Tianjin-Hebei, the carbon reduction pathway in the Pearl River-West River Economic Belt should consider regional heterogeneity and explore how policy spillover effects can promote coordinated emission reductions in pilot and non-pilot regions. Finally, in 2020, Guangdong and Guangxi within the economic belt achieved moderate and high levels of synergy in pollution and carbon reduction, respectively, indicating that policy coordination effects in the region have begun to emerge[22]. However, research on the spillover effects of policies in non-pilot areas within this region remains insufficient, highlighting the need for further exploration of the emission reduction effects of carbon trading policies both inside and outside pilot areas.

In summary, regarding the pollution and carbon reduction issues of the CTP, goal coordination mainly focuses on the relationship between a single pollutant and carbon emissions, and the conclusions on emission reduction synergy are highly controversial. Regional coordination pays little attention to the spatial spillover effects of carbon trading policies in regions at specific development stages, neglecting the issue of regional policy heterogeneity. Although the mechanism focuses on pollution and carbon reduction, it lacks an analysis of the heterogeneity of the "pollution reduction" and "carbon reduction" mechanisms. Based on this, the study has three marginal contributions: First, from the dual perspectives of goal coordination and regional coordination, it explores the role of the CTP in the pilot areas of the Pearl River-Xijiang Economic Belt and its spatial spillover effects on non-pilot areas. Second, it investigates the heterogeneity of the implementation effects of the CTP from the perspectives of regional differences and the dual aspects of "pollution reduction" and "carbon reduction." Third, from the comprehensive process perspectives of "source prevention," "process control," and "end treatment," it systematically examines the specific mechanisms of the CTP's effects on carbon emissions and pollutants through direct and indirect effects.

2.CTP Policy Impact Mechanism

The CTP is derived from Coase's property rights theory, which posits that if property rights can be clearly defined and transaction costs are zero, the market can achieve Pareto's optimal resource allocation through free trade. Under the influence of the Coase theorem, CTP has been widely applied to address environmental externalities. Numerous studies have shown that CTP not only has a significant effect on carbon reduction[10] but also promotes the reduction of pollutants due to the common source of carbon and pollution[24]. However, the free flow of regional

resources and elements not only affects the pilot areas themselves but also impacts surrounding areas[26]. Within the framework of the CTP, pilot areas achieve their environmental governance goals through pollution and carbon reduction, and also bring broader environmental benefits through spillover effects[27]. Based on the above analysis, considering that the Pearl River-Xijiang Economic Belt has the support of regional development policies, close economic development ties among cities, and free flow of resources and elements, we propose Hypotheses 1 and 2 :

H1: The CTP has facilitated the reduction of pollution and carbon emissions levels in the pilot areas of the Pearl River-Xijiang Economic Belt.

H2: The CTP has promoted the decline in pollution and carbon emissions levels in the Pearl River-Xijiang Economic Belt through the spatial spillover effect.

The Porter Hypothesis proposes the innovation offset effect and compliance cost effect to explain the mechanisms of environmental regulation policies. This has further led some scholars to use factors such as industrial structure, energy efficiency, and technological innovation as alternative indicators for measuring the Porter effect.

From the perspective of source prevention, the industrial structure's tilt towards heavy industry has become a significant reason for the increase in carbon emissions in China. **Error! Reference source not found.** The market plays a decisive role in carbon emission trading. Based on the "compliance cost" theory and the "profit maximization" principle[12], if the revenue from selling carbon emission allowances exceeds the innovation cost of green low-carbon technology, a large amount of capital will flow into the green low-carbon industry. This will lead to the gradual elimination of "high pollution, high emission, and high energy consumption" backward industries and the vigorous development of new industries with low pollution, low emission, and high technology[29]. Consequently, this promotes the transformation and upgrading of the industrial structure in pilot provinces and cities from low-end to high-end, effectively reducing carbon and pollution emissions.

From the perspective of "process control," energy consumption is the root source of CO₂ and pollutant emissions. In this context, all sectors are driven to improve production processes, advance production technology, and enhance energy efficiency to meet future market demands and regulatory requirements, thereby reducing levels of pollution and carbon emissions[30]. Additionally, to meet the emission targets set by the CTP, companies will also actively take measures such as substituting high-carbon energy like coal with clean energy, including wind, hydro, and solar power, to reduce coal energy consumption and further promote reductions in pollution

and carbon levels.

The CTP can stimulate pilot enterprises to engage in low-carbon technological innovation activities through the "signal-expectation" mechanism[19]. From the perspective of "end treatment," the CTP will to some extent increase pollution control costs for enterprises. To maintain operations, enterprises will increase investment in innovation and research, improving processes through technological innovation to reduce air pollution and carbon emission[31]. Due to population mobility, surrounding cities will imitate the emission reduction technologies of pilot cities, thereby driving pollution and carbon reduction efforts in surrounding cities[15]. This imitation effect of technological innovation has resulted in a "1+1>2" carbon reduction effect. Based on the above analysis, the study constructs an analytical framework for the CTP mechanism (Figure 1) and proposes three hypotheses for its impact pathways:

H3: The CTP has contributed to the reduction of pollution and carbon levels in the Pearl River-Xijiang Economic Belt by advanced industrial structure.

H4: The CTP has facilitated the reduction of pollution and carbon levels in the Pearl River-Xijiang Economic Belt by enhancing energy efficiency.

H5: The CTP has promoted the reduction of pollution and carbon levels in the Pearl River-Xijiang Economic Belt by fostering technological innovation.

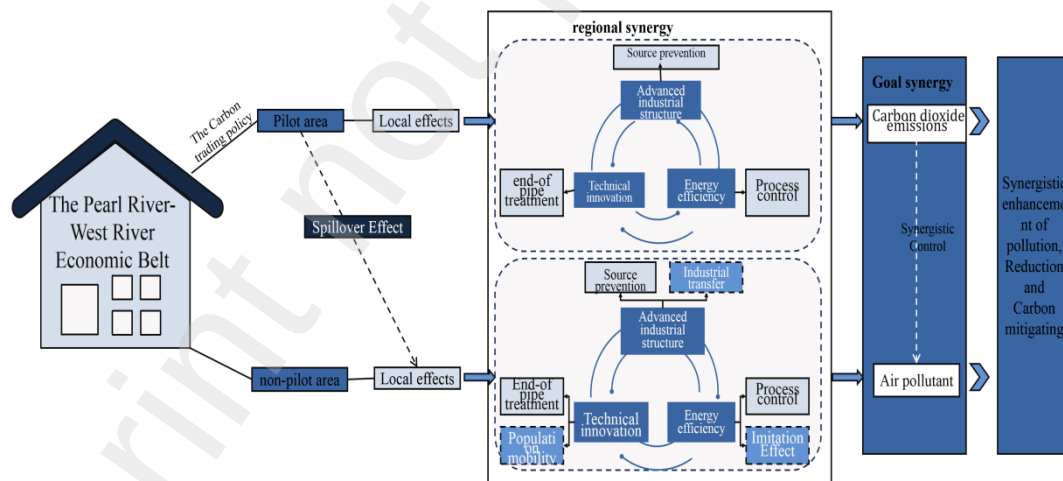


Fig. 1 Policy impact mechanism

3. Research design and data sources

3.1 Research method

The research methodology design involves first using the DID approach to evaluate the policy effects in the pilot cities within the economic belt, then using the

synthetic control method for robustness checks, followed by using spatial S-DID to explore the spatial spillover benefits of the policy in the economic belt, and finally employing the mediation effect model to investigate the mechanism of the policy.

The core idea of DID is to evaluate the impact of policy changes on the studied variables by comparing the experimental group and the control group. The DID model is widely used to assess the impact of the CTP on pollution reduction. By comparing the experimental group and the control group before and after the implementation of the CTP, the actual effect of the CTP on pollution and carbon reduction levels can be determined. The DID model constructed for this study is as follows:

$$Y_{it} = \beta_0 + \beta_1 Treated_i \times Time_t + \beta_2 Treated_i + \beta_3 Time_t + \sum \rho Control_{it} + \gamma_t + \lambda_i + \varepsilon_{it} \quad (1)$$

Y_{it} represents the dependent variables, which are measured by the levels of pollution and carbon reduction, CO₂ emissions, SO₂ emissions, total particulate matter, and PM_{2.5} concentration. The level of pollution and carbon reduction is represented by the interaction term of carbon emissions and the environmental pollution index. The core explanatory variable is the net effect of the CTP ($Treated \times Time$), where $Treated_i$ is the dummy variable for each city and $Time_t$ is the dummy variable. In the economic belt, Guangzhou, Foshan, Zhaoqing, and Yunfu officially launched the CTP in December 2013. Based on this, 2014 is considered the policy implementation time. $Treated \times Time$ is set to 1 if the city implements the policy, and 0 otherwise. $Control_{it}$ represents the control variables, γ_t represents the time-fixed effects, λ_i represents the city-fixed effects, and ε_{it} represents the random error term.

The synthetic control method assumes there are $J+1$ regions under study, with the first region being the policy-affected area, and the remaining J regions serving as potential control groups. Let Y_{it}^N denote the outcome variable for region i ($i=1,2,...,J+1$) at time t ($t=1,2,...,T$) before policy implementation, and let T_0 represent the period before policy implementation. Let Y_{it}^I denote the outcome variable for region i at time t ($T_0+1 < t < T$) under policy influence. Assuming there is no effect of intervention before policy implementation on the outcome variable for all regions i at the time $t \in \{1,2,...,T_0\}$, we have $Y_{it}^N = Y_{it}^I$ for all regions i at the time $Y_{it}^N = Y_{it}^I$. Let $\alpha_{it} = Y_{it}^I - Y_{it}^N$ represent the policy intervention effect for region i at time t , then the

outcome variable for region i at time t is given by:

$$Y_{it} = Y_{it}^N + \alpha_{it} \quad (2)$$

$$Y_{it}^N = \delta_t + \theta_t Z_i + \lambda_t \mu_i + \varepsilon_{it} \quad (3)$$

Considering the weight column vector of the first order J of which the region $W^* = (\omega_2^*, \dots, \omega_{J+1}^*)$ $j = 2, \dots, J+1$. The weight vector W^* , each element represents ω_j , and $\omega_2 + \dots + \omega_{J+1} = 1$. Each element of the weight column vector W^* represents a potential synthetic control value, namely the specific weighted average value of the control group regions. Introducing Equation (4), the estimated value of Y_{it}^N is:

$$Y_{it}^N - \sum_{j=2}^{J+1} \omega_j^* Y_{jt} = \sum_{j=2}^{J+1} \omega_j^* \sum_{s=1}^{T_0} \omega_j^* \lambda_s \left(\sum_{i=1}^{T_0} \lambda_i' \lambda_i \right)^{-1} \lambda_s' (\varepsilon_{js} \varepsilon_{is}) - \sum_{j=2}^{J+1} \omega_j^* (\varepsilon_{jt} - \varepsilon_{it}) \quad (4)$$

Existing studies have shown that the right-hand side of Equation (4) tends to approach 0. $\sum_{j=2}^{J+1} \omega_j^* Y_{jt}$ Y_{it}^N is an unbiased estimator of Y_{it}^N , $\alpha_{it} = Y_{it} - \sum_{j=2}^{J+1} \omega_j^* Y_{jt}$ is an unbiased estimator of α_{it} .

The Spatial Durbin Model (SDM) incorporates spatial lag terms of both dependent and independent variables, capturing spatial interactions and dependencies between regions. It reflects the impact of local and neighboring area variables on local dependent variables, aiding in the understanding of spatial associations and propagation mechanisms. SDM combines the advantages of the Spatial Autoregressive Model (SAR) and the Spatial Error Model (SEM) for problem analysis. Based on model (1) and the research of Yang^{Error! Reference source not found.}, the model is set as follows:

$$Y_{it} = \beta_0 + \alpha W \times Y_{it} + \beta_1 W Treated_i \times Time_t + \beta_2 Treated_i + \beta_3 Time_t + \sum \rho Control_{it} + \lambda W + \gamma_t + \lambda_i + \varepsilon_{it} \quad (5)$$

Where, W represents standardized spatial inverse distance matrix constructed based on the geographical locations of cities, α representing the spatial autoregressive coefficient of the dependent variable. β_2 denotes the spillover effect of the policy, while λ represents the spillover effect of the control variables.

To further explore the mechanism through which CTP achieves pollution reduction and carbon reduction in the Pearl River-Xijiang Economic Belt, we refer to Baron's approach and construct a mediation model to examine hypotheses H3 to H5.

$$M_{it} = \beta_0 + \beta_4 Treated_i \times Time_t + \beta_2 Treated_i + \beta_3 Time_t + \sum \rho Control_{it} + \gamma_t + \lambda_i + \varepsilon_{it} \quad (6)$$

$$Y_{it} = \beta_0 + C_1 M_{it} + \beta_2 Treated_i + \beta_3 Time_t + \sum \rho Control_{it} + \gamma_t + \lambda_i + \varepsilon_{it} \quad (7)$$

$$Y_{it} = \beta_0 + \beta_5 Treated_i \times Time_t + C_2 M_{it} + \beta_2 Treated_i + \beta_3 Time_t + \sum \rho Control_{it} + C_1 \gamma_t + \lambda_i + \varepsilon_{it} \quad (8)$$

Where, M_{it} represents the mediating variables, including industrial structure upgrading, energy efficiency, and technological innovation. First, perform regression on the model (6). If β_4 is significant, it indicates that the policy affects the mediating variables. Then, perform regression on the model (7). If C_1 is significantly negative, it indicates that the mediating variables reduce Y_{it} . Finally, perform regression on the model (8) and compare β_3 and β_5 . If β_5 is smaller than β_3 or its significance level decreases, it indicates the presence of a mediating effect, meaning that the carbon trading policy influences Y_{it} through the mediating variables.

3.2 Data source and variables selection

The Pearl River-Xijiang Economic Belt spans Guangdong and Guangxi, covering 11 prefecture-level cities and extending approximately 1,400 kilometers, as shown in Figure 2. This economic belt boasts abundant natural resources and a strong ecological foundation. The levels of pollution and carbon reduction are influenced by various factors, including regional resource and environmental carrying capacity, the economic and social development foundation, and the advancement of green policies.

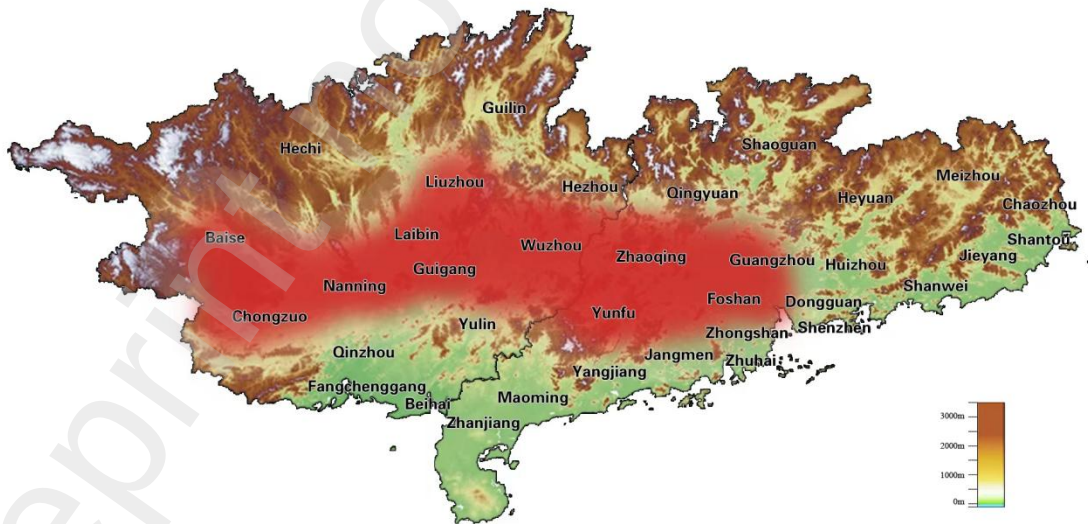


Fig. 2 Study area

Considering existing research, an evaluation system for pollution and carbon reduction levels was constructed using the interaction term of carbon emissions and the environmental pollution index. This interaction term can reflect the common

origin characteristics, aligning with the "carbon and pollution co-origin" theory. To calculate the environmental pollution index, emission indicators such as CO₂, SO₂, total particulate matter, and PM_{2.5} concentrations were selected and calculated using the entropy method. This index is a reverse indicator. Referring to related research, weather, economy, and environment are selected as control variables, using panel data from 2008 to 2021 in the economic belt. Data sources and variable indices are shown in Table 1. Since the National Bureau of Statistics changed the statistical criteria for total particulate matter in 2011, the annual total particulate matter is calculated by summing the emissions of smoke and industrial dust. To eliminate heteroscedasticity, except for ratios and dummy variables, all other variables are log-transformed. Per capita GDP is adjusted for inflation to 2021 constant prices.

Table 1 Description of variables and data sources

type	Variable name	definition	data sources
explained variable	Reduce pollution and reduce carbon level	Log of pollution reduction and carbon reduction index	China Carbon Accounting database
	CO ₂ emissions	Log of CO ₂ emissions	China Carbon Accounting Database
	SO ₂ emissions	Log of SO ₂ emissions	The China Environmental Statistics Yearbook
	Total emission of smoke and dust	Log of smoke and dust emissions	The China Environmental Statistics Yearbook
	And the PM _{2.5} concentration	Log of PM _{2.5} emissions	Research team at Washington University, St. Louis, USA
controlled variable	Weather[32]	mean annual precipitation	China Hydrology Annual Report
		mean wind speed	The China Meteorological Data Network
		mean annual temperature	The China Meteorological Data Network
	economy	Population density	The Guangdong and Guangxi Statistical Yearbook
		Green coverage rate of urban built-up areas	The Guangdong and Guangxi Statistical Yearbook
		Per capita GDP	CEIC database
		The added value of the	The Guangdong and Guangxi

		secondary industry accounted for GDP	Statistical Yearbook
	environment	green finance index	CEIC database
		The number of buses in possession	CEIC database
		The authorized quantity of green patents	CEIC database
Mechanism variables	Advanced industrial structure[30]	From the perspective of "quality", "quantity" and measure the advanced industrial structure (TS)	The Guangdong and Guangxi Statistical Yearbook
	energy efficiency[33]	Log value of energy saving and emission reduction efficiency	The China Energy Statistical Yearbook
	technical innovation	Number of patents per 100 people	CEIC database

4. Empirical analysis

4.1 Characterizing factual descriptions

First, an assessment of the pollution and carbon reduction levels in the Pearl River-Xijiang Economic Belt during the study period was conducted to explore its changing characteristics, as shown in Figure 3. Figure 3 shows that the pollution and carbon reduction levels in the economic belt generally show a downward trend, with the lowest level occurring in 2015, the second year of policy implementation. The reduction level in pilot cities is significantly lower than that in non-pilot areas. Is the reduction in pollution and carbon levels related to the CTP? Through what mechanisms is this promoted? Are the mechanisms for "pollution reduction" and "carbon reduction" the same? These questions will be explored through empirical analysis in the following sections.

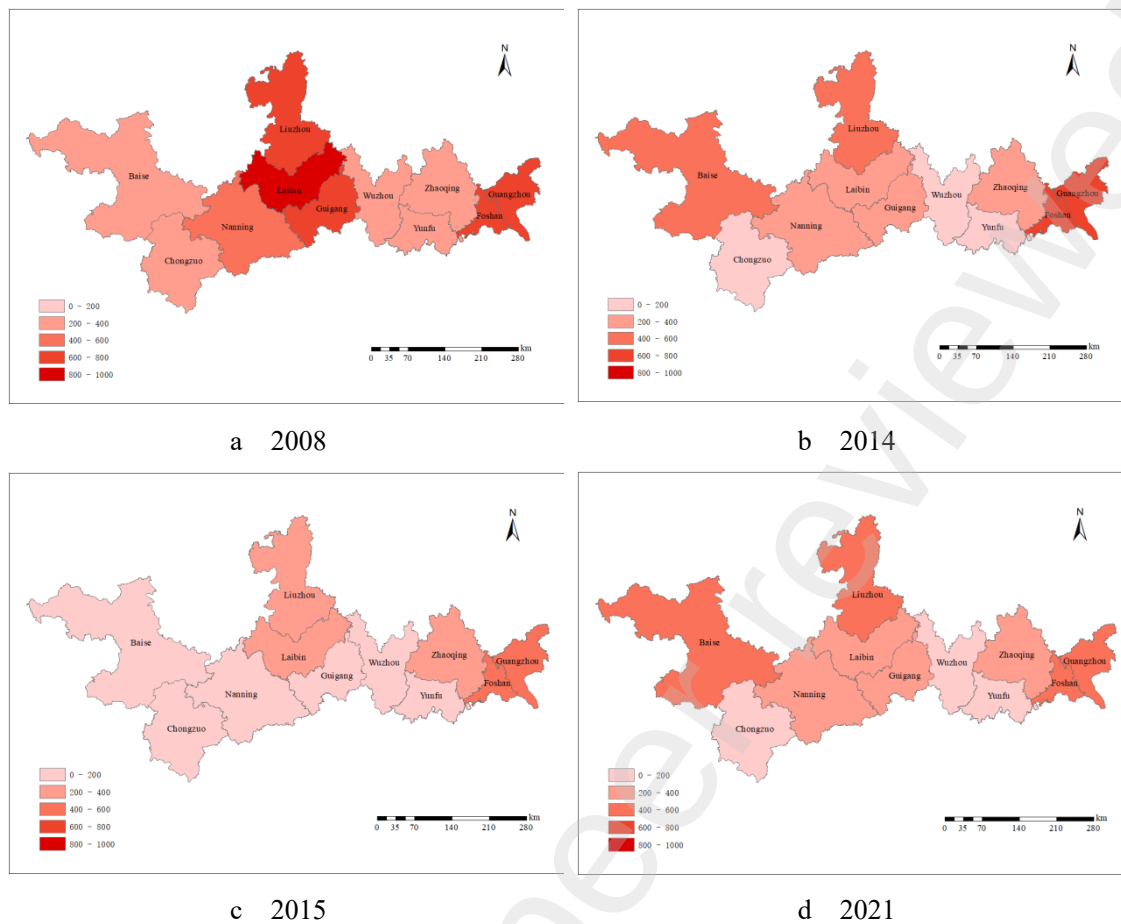


Fig. 3 Pollution reduction and carbon reduction level

4.2 Policy effect assessment

Table 2 shows the results of the difference-in-differences (DID) analysis. Model 1 is the baseline model, which does not consider any control variables, while Models 2 to 7 progressively introduce control variables. The results indicate that, regardless of whether control variables are considered, the dummy variable for the CTP is significantly negative at the 1% level. This implies that the CTP promotes pollution and carbon reduction in the pilot areas, reducing levels by 34.1%. Therefore, based on the results of the baseline regression, Hypothesis 1 (H1) is validated. However, further tests are needed to examine its endogeneity and robustness.

Table 2 DID regression results

variable	model 1	model 2	model 3	model 4	model 5	model 6	model 7
DID	-0.408**	-0.396**	-0.388**	-0.379**	-0.366**	-0.352**	-0.341***
	*	*	*	*	*	*	(-0.140)
	(-0.126)	(-0.081)	(-0.090)	(-0.106)	(-0.117)	(0.113)	
weather		-0.154**	-0.143*	-0.152**	-0.147**	-0.149**	-0.151**
		(0.079)	(0.067)	(0.074)	(0.081)	(0.068)	(0.071)

density of			1.982*	1.879**	1.673**	1.792**	1.8112**
population			(2.416)	(2.383)	(2.181)	(2.301)	(2.056)
Green							
coverage				-0.671**	-0.642**	-0.595**	
rate of				*	*	*	-0.595***
urban				(-1.897)	(-1.786)	(-1.649)	(-1.649)
built-up							
areas							
per capita					0.073***	0.067***	0.069***
GDP					(0.127)	(0.114)	(0.125)
The added							
value of							
the							
secondary						0.429***	0.339***
industry						(5.397)	(4.511)
accounted							
for GDP							
environme							-1.021***
nt							(-2.511)
constant	3.239***	3.224**	6.531**	4.218***	5.661**	-8.990**	5.367**
term	(0.024)	(0.780)	(0.667)	(0.987)	(0.089)	(5.539)	(2.211)
observed							
value	154	154	154	154	154	154	154
controlled							
variable	no	3	4	5	6	7	10
time							
effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Location							
fixed	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: Values in parentheses are standard errors. *, ** and *** indicates the significant levels of 10%, 5% and 1 %, respectively

4.2.1 Parallel trend test

DID model must meet the parallel trends assumption, where the pilot and non-pilot areas follow similar trends before the policy implementation. This study uses the event study method to test the parallel trends assumption. The horizontal axis

from -4 to -1 represents the dummy variables for the four years before policy implementation, 0 represents the dummy variable for 2014, and 1 to 5 represents the dummy variables for the five years after policy implementation. The vertical axis represents the coefficient estimates. The coefficients for pollution and carbon reduction levels are not significant in the four years before the implementation of the CTP, indicating no significant systematic differences between the pilot and non-pilot areas before the policy was implemented, thus meeting the parallel trends assumption. Additionally, the estimated coefficients after policy implementation indicate that the CTP has a sustained effect on promoting pollution and carbon reduction in the pilot areas.

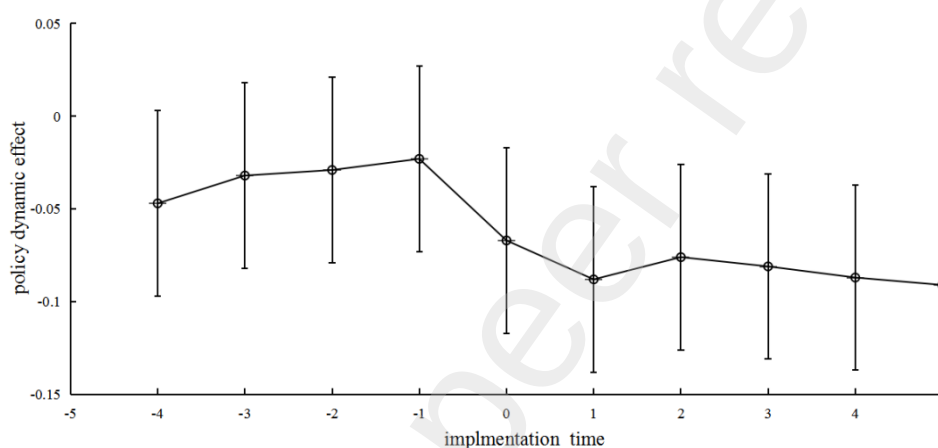


Fig. 4 Parallel trend test through difference-in-difference model

4.2.2 Endogenous test

To further test the robustness of the results and overcome the sample selection bias and endogenous issues caused by unobserved factors in the traditional DID model, this paper adopts the SCM[34]. The study uses control variables from 2008 to 2021 as predictor variables and calculates the synthetic weights for each carbon trading pilot city using the "synth" command in Stata16. The specific results are shown in Table 3. By summing the weights of each city, we can estimate the pollution and carbon reduction levels of each pilot city before the CTP was implemented.

Table 3 Synthetic weights of the pilot areas

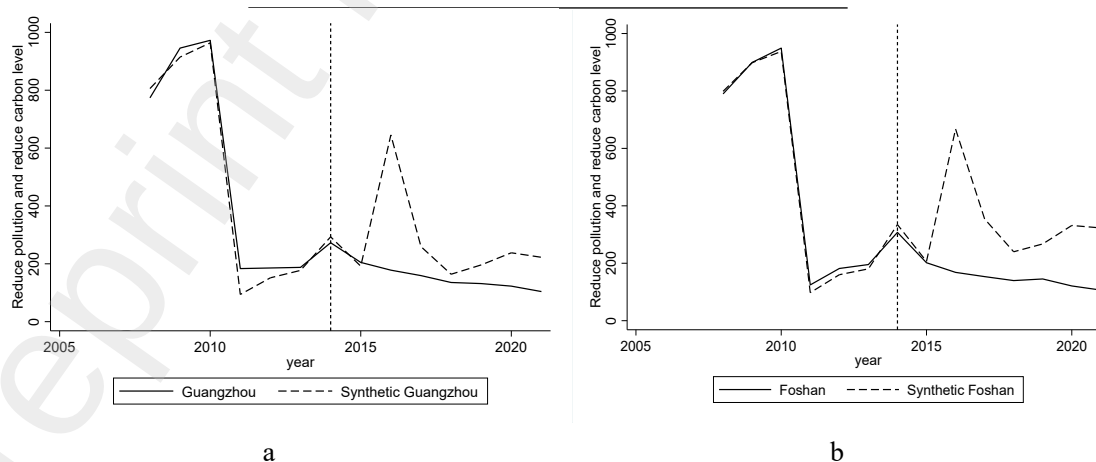
City	Nanning	Liuzhou	Wuzhou	Guigang	Baise	Laibin	Chongzuo
Guangzhou	0.083	0	0	0.876	0	0.041	0
Foshan	0	0.194	0	0.766	0	0.040	0
Zhaoqing	0.399	0.039	0.555	0.007	0	0	0
Yunfu	0	0	0	0.015	0.202	0	0.783

Figure 5 shows the fitting results of the pollution and carbon reduction levels in

each pilot city before and after the implementation of the CTP under the SCM model. The vertical dashed line represents the policy implementation time in the pilot cities, while the horizontally varying dashed and solid lines represent the synthetic and actual values of the pollution and carbon reduction levels in the pilot cities, respectively. By observing Figure 5 and calculating the root mean squared prediction error (RMSPE) values for each pilot city (Table 4), it can be seen that before the policy implementation, the actual values and synthetic values of the pilot cities show roughly the same trend. After the policy implementation, the actual values are significantly lower than the synthetic values, further validating Hypothesis H1. Overall, the actual pollution and carbon reduction levels show a linear downward trend, while the synthetic values show a fluctuating upward trend. The two gradually diverge, and the difference between them gradually increases. In terms of individual policy effects in the pilot cities, Zhaoqing performs relatively well, while the pollution and carbon reduction levels in Guangzhou and Foshan have decreased but remain at relatively high levels. This individual difference may stem from varying responses to the CTP in different cities, involving factors such as each city's unique industrial structure, pollution source distribution, and specific details of policy implementation [22].

Table 4 Square root calculation results of mean square prediction error

Pilot cities	RMSPE
Guangzhou	0.011
Foshan	0.006
Zhaoqing	0.014
Yunfu	0.012



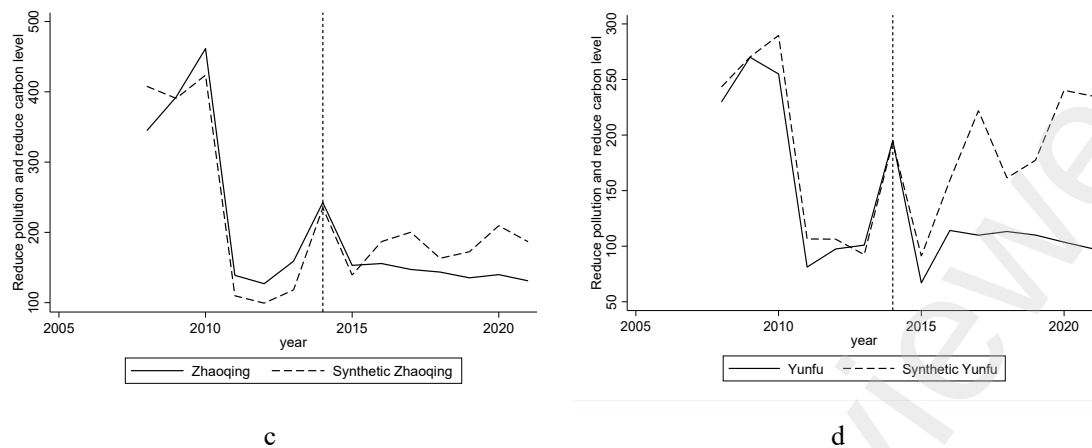


Fig. 5 Endogeneity Test Results

4.2.3 Robustness tests

To further test the robustness of conclusion H1, this study adopts the methods of a counterfactual test, eliminating the interference of other policies, and shortening the time window for the test. The results are shown in Table 5. First, the counterfactual test is carried out by investigating the impact of DID on carbon emission intensity when the carbon trading policy is not implemented, to verify the research conclusion. Column (1) indicates that the estimated coefficient is no longer significant, which means that there are no other systematic errors in the experimental group and the control group, and the benchmark regression result is credible. During the research period, the level of pollution reduction and carbon reduction may be impacted by other similar policies, such as the opening of the national carbon market in July 2021. To obtain the net effect of the carbon trading pilot policy, this study incorporates this policy into the regression analysis to eliminate the interference of other policies. Column (2) shows that after excluding the opening policy of the national carbon market, the net effect of the policy is consistent with the conclusion of the benchmark regression. This may be ascribed to the fact that during the incipient phase of the unified market's formation, the market mechanism remains incomplete and only the power industry is involved. Simultaneously, it is also subject to the influence of the macro-economy and the intrinsic attributes of the industry. Finally, considering that the selection window of sample time may influence the estimation of policy effects, this study narrows the sample time window to 2010 - 2019. As shown in Column (3), the coefficient before the DID is still significantly negative, from which it can be inferred that the main regression result is of a certain robustness.

Table 5 Robustness Test Regression Results

explanatory	explained variable
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variable	(1)counterfactual	(2)eliminate other policy interference	(3)shortening the time window
DID	-0.178 (0.391)	-0.359*** (0.279)	-0.137*** (0.027)
constant term	-3.998*** (1.523)	-4.326* (0.786)	-2.963*** (1.779)
Adjust the R ²	0.962	0.953	0.981
observed value	154	154	56
controlled variable	Yes	Yes	Yes
time effect	Yes	Yes	Yes
Individual effect	Yes	Yes	Yes

Note: Values in parentheses are standard errors. *, ** and *** indicates the significant levels of 10%, 5% and 1 %, respectively

4.3 Analysis of the spatial effects

First, the spatial autocorrelation method is used to explore the spatial correlation of pollution reduction and carbon reduction in the region (Appendix Table 1 for details). It is found that the Pearl River-Xijiang Economic Belt exhibits spatial spillover effects and a strong positive clustering trend in pollution and carbon reduction. Based on this, the S-DID method is attempted. Table 6 presents the results of the S-DID, which indicate that the direct effect of the carbon trading pilot policy on carbon intensity is greater than the indirect effect, and both effects are significantly negative. Model 8 is a two-way fixed-effects spatial Durbin model without control variables; model 9 is a time-fixed-effects model with control variables; model 10 is an individual fixed-effects model with control variables; and model 11 is a two-way fixed-effects spatial Durbin model with control variables. In all models, the dummy variable is significantly negative at the 1% level, and the R-sq of each model indicates a good fit. Therefore, it can be concluded that the carbon trading pilot policy has a spatial spillover effect, achieving emission reductions not only in the pilot cities but also reducing the pollution reduction and carbon reduction levels of the economic belt by 29.9%, thus confirming hypothesis H2. Specifically, the pilot areas' governments have strengthened effective inspection and governance of pollution and carbon

reduction, setting a model for neighboring areas and significantly promoting regional pollution and carbon reduction levels. Meanwhile, to promote pollution and carbon reduction, the flow of factors within the economic belt has become more frequent, resulting in the spillover of low-carbon technological knowledge, providing external drive and internal support for neighboring areas to reduce carbon intensity.

Table 6 The robustness test results of S-DID

variable	model 8	model 9	model 10	model 11
DID	-0.321*** (-0.214)	-0.308*** (-0.127)	-0.311*** (-0.198)	-0.299*** (-0.187)
α	0.211*** (0.071)	0.237*** (0.075)	0.208*** (0.070)	0.139** (0.034)
w	0.007*** (0.000)	0.040*** (0.003)	0.005*** (0.000)	0.005*** (0.000)
direct effect	-0.322*** (-0.214)	-0.385*** (-0.153)	-0.313*** (-0.198)	-0.311*** (-0.187)
indirect effect	-0.025 (0.056)	-0.077*** (-0.123)	-0.084* (0.048)	-0.044* (0.067)
Controlled variable	no	yes	yes	yes
time effect	yes	yes	no	yes
Location fixed	yes	no	yes	yes
R-sq	0.987	0.876	0.961	0.892
observed value	154	154	154	154

Note: Values in parentheses are standard errors. *, ** and *** indicates the significant levels of 10%, 5% and 1 %, respectively

4.4 Heterogeneity test

There are significant differences among cities in the Pearl River-Xijiang River Economic Belt in terms of policy implementation, industrial structure, resource carrying capacity, and economic development level. These differences may lead to heterogeneity in the pollution reduction and carbon reduction effects of the CTP.

According to the "Blue Book of the Pearl River-Xijiang River Economic Belt" and the "Development Plan of the Pearl River-Xijiang River Economic Belt" issued by the Central Committee of the Communist Party of China and the State Council in 2014, this economic belt is divided into four groups: Guangzhou-Foshan, Nanning-Chongzuo-Baise, Zhaoqin-Yunfu-Guigang-Wuzhou, and Liuzhou-Laibin. Further

parameter estimation for the above four sub-samples is shown in Table 7. Table 7 indicates significant regional heterogeneity in the pollution reduction and carbon reduction levels due to the CTP in the Pearl River-Xijiang Economic Belt. The CTP's pollution and carbon reduction levels are significantly negative at the 1%, 5%, and 5% levels in Guangzhou-Foshan, Nanning-Chongzuo-Baise, and Zhaoqing-Yunfu-Guigang-Wuzhou, respectively. In Liuzhou-Laibin, the levels are negative but not significant. This result also indicates that regions with higher levels of economic development and market activity provide suitable conditions for the implementation of carbon trading policies. The CTP in Nanning-Chongzuo-Baise has a significantly negative effect on pollution and carbon reduction at the 5% level, which can be attributed to Nanning's pioneering carbon account initiatives and active participation in the "dual carbon" policy system. The CTP's pollution and carbon reduction levels in Liuzhou-Laibin are negative but not significant, as both cities have high energy consumption bases, economic development sensitive to energy demand, and limited potential for energy savings and carbon reduction in key industries.

Table 7 Regional heterogeneity

project	Guangzhou-Foshan	Nanning-Chongzuo-Baise	Zhaoqing-Yunfu-Guigang-Wuzhou	Liuzhou-Lai bin
DID	-0.468*** (-0.341)	-0.301** (-0.086)	-0.347** (-0.161)	-0.311 (-0.108)
constant term	3.998*** (1.523)	4.326*** (0.786)	2.963*** (1.779)	3.523*** (2.381)
adjust R ²	0.962	0.953	0.981	0.927
observed value	154	154	154	154
controlled variable	10	10	10	10
time effect	Yes	Yes	Yes	Yes
Location fixed	Yes	Yes	Yes	Yes

Note: Values in parentheses are standard errors. *, ** and *** indicates the significant levels of 10%, 5% and 1 %, respectively

To further verify the impact of the CTP on "pollution reduction" and "carbon reduction," this study employs S-DID to empirically test the emissions of CO₂, SO₂, dust and particulate matter, and PM_{2.5}. In this analysis, odd-numbered columns do not

consider control variables, while even-numbered columns include them. Table 8 shows that, regardless of whether control variables are added, the implementation of the policy has a significant reduction effect on CO₂ and the three air pollutants. The even-numbered columns of Table 8 indicate that, after adding control variables, the explanatory power of the models significantly increased. The coefficients of the policy implementation variables are less than 0 at significance levels of 0.01 and 0.05, indicating that carbon trading policies significantly reduce CO₂ and air pollutant emissions. Specifically, CO₂ emissions decreased by 28.7%, SO₂ emissions decreased by 30.9%, and dust emissions decreased by 23.2%. The policy's effect on PM_{2.5} reduction is not significant, with a decrease of 3.1%. This may be because PM_{2.5} is a secondary pollutant with a complex composition of sources, and coal combustion contributes less to its total emissions. The synergistic reduction effect of CO₂ and SO₂ is the best, likely because coal remains the primary energy source in China, and the sulfur content in Chinese coal is relatively high.

Table 8 Regression results of pollution reduction and carbon reduction heterogeneity

variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CO ₂	CO ₂	SO ₂	SO ₂	Total amount of smoke and dust	Total amount of smoke and dust	PM _{2.5}	PM _{2.5}
DID	-0.373* ** (-6.216)	-0.287** * (-2.854)	-0.398* ** (-7.454)	-0.309* ** (-5.709)	-0.245*** (-3.354)	-0.232*** (-3.001)	-0.076* ** (-1.531)	-0.031 ** (-1.001)
controlled variable	No	yes	no	yes	no	yes	no	yes
Time fixed	yes	yes	yes	yes	yes	yes	yes	yes
Location fixed	yes	yes	yes	yes	yes	yes	yes	yes
Observations	154	154	154	154	154	154	154	154
R-squared	0.880	0.998	0.876	0.889	0.993	0.971	0.890	0.897

Note: Values in parentheses are standard errors. *, ** and *** indicates the significant levels of 10%, 5% and 1 %, respectively

4.5 Mechanism of action analysis

Summarizing the relevant literature, this study explores the mechanisms of carbon trading policies and systematically considers the entire process of carbon and pollutant emissions, focusing on three pathways: industrial structure upgrading, energy efficiency, and technological innovation[30]. The study decomposes the impact of each mechanism variable into direct effects and spillover effects. The direct effects indicate the impact on pollution and carbon reduction in the policy implementation area, while the spillover effects represent the spatial spillover effects on surrounding areas. Table 9 presents the regression results of the impact mechanisms.

Column (1) shows that the direct effect of the CTP on upgrading the industrial structure is significantly positive, while the spillover effect is significantly negative, resulting in an overall negative effect. This indicates that the CTP in the current economic belt has not generally promoted industrial structure upgrading, thus hypothesis H3 is not verified. The reasons for this include: first, the CTP at the pilot stage covers only a limited number of industries, having a relatively weak impact on the overall industrial structure. Second, post-policy implementation, the Guangdong section experienced a "siphon effect," causing resource loss in surrounding areas, and the industrial structure adjustment triggered by carbon trading negatively impacted traditional industries in neighboring regions. Finally, due to financial and technological constraints, enterprises tend to adopt short-term, low-cost emission reduction measures, without fundamentally altering the industrial layout.

Column (3) shows that the direct and spillover effects of the CTP on energy efficiency are significantly positive. Column (4) indicates that after including energy efficiency in the model, the coefficient of policy implementation is significantly negative, decreasing by 0.048. Therefore, hypothesis H4 is confirmed, indicating that the CTP reduces the overall pollution and carbon reduction level of the economic belt by improving energy efficiency. These results can be explained by the fact that the CTP accelerates the elimination of low-energy-efficiency production units within the economic belt, altering the dilemma of being locked in the low end of the value chain. The improvement in energy efficiency effectively reduces the region's pollution and carbon reduction levels. A well-developed market mechanism helps to stimulate the vitality of market entities, break down regional interest barriers, and promote the free flow of resources and factors, thereby facilitating the reduction of pollution and carbon levels in local and neighboring cities.

Column (5) shows that the direct and spillover effects of the CTP on technological innovation are significantly positive, with the spillover effect being greater than the direct effect. After including technological innovation in the model, the policy implementation coefficient is significantly negative, decreasing by 0.079, indicating that the CTP reduces pollution and carbon levels through technological innovation, confirming hypothesis H5. The above results indicate that the CTP has distinct factor-biased characteristics, with technological innovation understood as the embedding and integration of factors into the existing production processes of the economic belt. Technological innovation requires investment, which can improve the local environment and, through spatial spillover effects, allow neighboring regions to benefit without additional input. Additionally, the neighboring regions experience a "copycat effect," further promoting pollution and carbon reduction.

Table 9 Results of the mechanism of action regression

variable	industrial structure		energy efficiency		technological innovation	
	upgrading					
	(1)	(2)	(3)	(4)	(5)	(6)
S-DID	-0.020**	-0.306***	0.162***	-0.356***	0.193***	-0.387***
	(0.053)	(-2.941)	(2.053)	(-1.321)	(2.794)	(-1.989)
direct effect	0.101**	-0.338***	0.101***	-0.314***	0.070***	-0.171***
indirect effect	-0.121**	0.002**	0.051***	-0.042***	0.123***	-0.216***
controlled variable	Yes	Yes	Yes	Yes	Yes	Yes
Bootstrap test	0.113***		0.078***		0.097***	
Year fixed	Yes		Yes		Yes	
Location fixed	Yes		Yes		Yes	
constant term	0.254***	1.398***	0.554***	2.398***	6.987***	5.971***
	(4.385)	(6.617)	(7.383)	(11.697)	(10.273)	(14.076)
observed value	154	154	154	154	154	154
R ²	0.886	0.923	0.743	0.833	0.886	0.933

Note: Values in parentheses are standard errors. *, ** and *** indicates the significant levels of 10%, 5% and 1 %, respectively

To accurately identify the pathways through which carbon trading policies impact carbon and pollutant emission reductions, a mediation effect model was used,

with the explanatory variables set as CO₂, SO₂, particulate matter, and PM_{2.5}. Table 10 shows that, overall, the CTP has a significant and notable effect on reducing SO₂ emissions. Specifically, the pathways for reducing different pollutants vary. For CO₂ and PM_{2.5}, improving energy efficiency yields better reduction effects, as these pollutants originate from energy consumption. This method is direct, timely, and cost-effective. For SO₂, promoting industrial structure upgrading is the most effective reduction pathway, as it mainly originates from traditional heavy industries. Upgrading the industrial structure can control emissions at the source. For particulate matter, technological innovation is more effective in reducing emissions due to its diverse and dispersed sources, and reduction relies on advanced equipment.

Table 10 Results of regression of carbon reduction and pollution reduction mechanisms

variable	industrial structure upgrading		energy efficiency		technological innovation	
	(1)	(2)	(3)	(4)	(5)	(6)
S-DID (CO ₂)	-0.050** (-0.045)	-0.284*** (-2.561)	0.167*** (1.452)	-0.377*** (-1.761)	0.131*** (1.794)	-0.359*** (-1.874)
S-DID (SO ₂)	0.210*** (0.123)	-0.391*** (-2.331)	0.201*** (1.808)	-0.389*** (-2.121)	0.142*** (1.884)	-0.377*** (-1.981)
S-DID(particulate matter)	2.801* (1.810)	-0.301* (-2.109)	0.716** (2.531)	-0.308** (-2.114)	1.119*** (3.430)	-0.309*** (-1.909)
S-DID (PM _{2.5})	-0.325 (-1.275)	-0.298*** (-2.331)	0.336*** (2.751)	-0.393*** (-2.771)	-1.331** (-2.204)	-0.281*** (-1.543)
Year fixed	Yes		Yes		Yes	
Location fixed	Yes		Yes		Yes	
constant term	0.254*** (4.385)	1.398*** (6.617)	0.554*** (7.383)	2.398*** (11.697)	6.987*** (10.273)	5.971*** (14.076)
observed value	154	154	154	154	154	154
R ²	0.886	0.923	0.743	0.833	0.886	0.933

Note: Values in parentheses are standard errors. *, ** and *** indicates the significant levels of 10%, 5% and 1 %, respectively

5. Conclusions and Recommendation

Based on panel data from 2008 to 2021 for the Pearl River-Xijiang Economic

Belt, SCM and S-DID models were used to explore the effects of carbon trading policies on pollution reduction and carbon reduction in the pilot areas of the Pearl River-Xijiang Economic Belt, as well as their spatial spillover effects on non-pilot areas. The regional heterogeneity and target heterogeneity of the policy's spatial effects were considered. The mechanisms of carbon trading policies on pollution reduction and carbon reduction were systematically examined from the perspectives of "source prevention," "process control," and "end-of-pipe treatment," considering both direct and indirect effects.

Three distinct conclusions can be drawn: (1) Carbon trading policies have promoted a 34.1% overall reduction in pollution and carbon levels in the pilot areas and have had a sustained effect. Additionally, through spatial spillover effects, they have reduced pollution and carbon levels in the economic belt by 29.9%. (2) The effect of carbon trading policies on pollution and carbon reduction is more pronounced in areas with better economic development and active carbon trading. They have reduced CO₂ emissions by 28.7%, SO₂ emissions by 30.9%, and particulate matter by 23.2%. The effect on PM_{2.5} reduction is not significant, with a decrease of 3.1%. (3) The spillover effect of carbon trading policies on technological innovation is greater than the direct effect, resulting in the most noticeable pollution and carbon reduction. However, the overall effect on industrial structure is negative, indicating a failure to promote pollution and carbon reduction. The mechanisms for reducing different pollutants vary: CO₂ and PM_{2.5} are linked to energy efficiency; SO₂ to industrial structure upgrading; and particulate matter to technological innovation.

Coordinated control can not only reduce governance costs but also avoid high-carbon lock-in. To better leverage the synergistic emission reduction effect of carbon trading policies and help policymakers explore a complementary, coordinated, and win-win path for pollution reduction and carbon reduction, the following policy recommendations are proposed:

First, establish a regional carbon-pollution collaborative governance mechanism to enhance pollution reduction and carbon mitigation effects. Based on the development characteristics of the Pearl River-West River Economic Belt, it is recommended to construct a unified "Carbon-Pollution Collaborative Governance Platform" to facilitate information sharing, data exchange, and joint governance projects within the region. Developed cities such as Guangzhou and Foshan should take the lead in implementing carbon trading policies, leveraging their advantages in capital, technology, and industrial upgrading to drive the coordinated development of

less developed areas like Guangxi, thereby reducing overall carbon emissions. Specifically, for heavily polluted cities such as Nanning and Guigang, regional collaboration can reduce governance costs and increase the vibrancy and effectiveness of the carbon trading market.

Second, implement differentiated emission reduction strategies and strengthen the ecological compensation mechanism. According to the development levels and industrial structures of cities in the Pearl River-West River Economic Belt, a tiered emission reduction strategy should be applied. In regions with active carbon trading and stronger economies, such as Guangzhou and Foshan, stricter emission reduction standards should be implemented, particularly for $PM_{2.5}$, combined with energy structure adjustments and technological upgrades. In less developed areas like Guangxi, an ecological compensation mechanism should be established to prevent environmental issues caused by uneven resource allocation or pollution transfer. Incentive policies such as tax reductions and subsidies should be introduced to encourage enterprises to participate in the carbon trading market and promote the application of green technologies.

Third, promote industrial upgrading and technological innovation to support green and low-carbon transformation. For traditional high-pollution industries within the Pearl River-West River Economic Belt, it is recommended to establish a "Green Low-Carbon Industrial Transition Fund" to particularly support the low-carbon transformation of highly polluting industries in the Guangxi region, reducing SO_2 and particulate matter emissions. Carbon trading policies should be used to stimulate technological innovation, encouraging enterprises to increase research and development of energy-saving and emission-reduction technologies. Especially in manufacturing hubs like Foshan, technological spillovers should drive industrial upgrading in other cities within the region. The government should establish incentive mechanisms to promote green retrofitting of high-energy-consuming enterprises, enhance energy efficiency, and achieve effective reductions in CO_2 and $PM_{2.5}$ emissions, thereby promoting a synergy between economic and environmental optimization.

Despite the contributions of the study, it has certain limitations that suggest further research. The data for the pilot policy implementation period was used as the research object, which is reasonable and feasible but did not exclude the impact of the national carbon market in the second half of 2021. Subsequent research should be based on national data to re-verify the effectiveness and spatial spillover effects of

carbon trading policies and conduct in-depth analysis. Additionally, the synergistic mechanism of pollution reduction and carbon reduction in CTP was not further clarified. Future research will introduce four policy tools: carbon price, carbon market activity, carbon market size, and penalties. From the three approaches of industrial structure upgrading, energy efficiency, and technological innovation, the discussion will focus on how carbon trading policies affect pollution reduction and carbon reduction through specific policy tools.

Appendix

Table 1 Global global autocorrelation results

Variables	I	E(I)	sd(I)	Z	p-value
2008	0.271	-0.1	0.219	1.692	0.045
2009	0.314	-0.1	0.228	1.812	0.035
2010	0.283	-0.1	0.225	1.698	0.045
2011	0.344	-0.1	0.221	2.007	0.022
2012	0.361	-0.1	0.221	2.086	0.018
2013	0.235	-0.1	0.22	1.521	0.064
2014	0.188	-0.1	0.226	1.273	0.071
2015	0.289	-0.1	0.222	1.747	0.041
2016	0.187	-0.1	0.231	1.243	0.007
2017	0.285	-0.1	0.227	1.699	0.045
2018	0.114	-0.1	0.229	0.934	0.075
2019	0.067	-0.1	0.23	0.725	0.084
2020	0.024	-0.1	0.224	0.552	0.029
2021	0.026	-0.1	0.227	0.555	0.089

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